Amirhesam Torkashvand

M.Sc. Student in Computer Engineering (Artificial Intelligence & Data Analytics), Politecnico di Torino

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PROFILE & THESIS OBJECTIVE

Master's student at Politecnico di Torino (expected graduation July 2027) with hands-on research experience in computational pathology, visual foundation models (DINOv2/v3, SAM), explainable and fair machine learning, and agentic LLM systems. Experienced across the research cycle, from data curation and baselines to ablations and reproducible open-source code.

Seeking: a full-time **external master's thesis** (about 6 months, from early 2027) in a research group at a European university or research institute, in **computer vision, medical image analysis or trustworthy AI**, carried out on site under the host supervisor with co-supervision from a Politecnico di Torino faculty member. Strongly motivated to continue towards a PhD.

RESEARCH INTERESTS

Medical Image Analysis & Computational Pathology • Visual Foundation Models & Self-Supervised Learning • Explainable & Trustworthy AI • Vision–Language Learning • Video Understanding • Agentic AI & RAG

EDUCATION

M.Sc. in Computer Engineering – Artificial Intelligence & Data Analytics Sep 2024 – Jul 2027 (expected)
Politecnico di Torino, Turin, Italy

- Relevant coursework: Advanced Machine Learning; Machine Learning in Applications.

M.Sc. in Artificial Intelligence (not completed) Sep 2023 – Jul 2024
Islamic Azad University, Central Tehran Branch, Tehran, Iran

- Withdrew to enrol at Politecnico di Torino. Courses: Machine Learning, Deep Learning, Big Data Analytics, Voice and Speech Recognition.

B.Sc. in Electrical Engineering – Electronics Sep 2018 – Jun 2023
Islamic Azad University, Central Tehran Branch, Tehran, Iran

SELECTED RESEARCH PROJECTS

Glomeruli Segmentation & Unsupervised Necrotization Profiling in Kidney Biopsies *Medical AI · Computer Vision*
Machine Learning in Applications, Politecnico di Torino (team of 4)

- Built a three-phase pipeline on 9 whole-slide kidney biopsy images (677 annotated glomeruli; patient-level 7/2 train/validation split), using Macenko stain normalisation, deterministic translation jittering and tissue-verified hard-negative tiles to counter batch effects, centring bias and false positives.
- Fine-tuned COCO-pretrained YOLOv8x-seg for glomerulus instance segmentation: **mask mAP@50 0.993** (mAP@50–95 0.757) and **mean Dice 0.83** (median 0.93) on 170 held-out patches.
- Profiled mask-pooled, slide-batch-corrected DINOv2 embeddings with K-Means, HDBSCAN, UMAP and t-SNE, finding that necrotization forms **one continuous healthy-to-sclerotic axis** (PC1) rather than discrete stages A–F, not explained by crop framing, glomerulus size or slide.
- Confirmed the finding across 8 backbone and masking configurations, including the pathology foundation models **UNI** and **H0-mini** (cluster–slide NMI ≤ 0.10).

Semantic Correspondence with Visual Foundation Models *Computer Vision · Foundation Models*
Advanced Machine Learning, Politecnico di Torino (team of 3) **Code ↗**

- Benchmarked DINOv2, DINOv3 and SAM (ViT-B) as training-free dense feature extractors on SPair-71k; DINOv2 led zero-shot at 51.9% PCK@0.10, against 20.4% for SAM.
- Fine-tuned the top DINOv2 block with an InfoNCE patch-matching loss, raising PCK@0.10 to **67.4% (+15.5 pp)**; windowed soft-argmax decoding lifted it to **69.1%**.
- Ablated LoRA (rank 4, $<1\%$ of parameters), mutual-nearest-neighbour (MNN) verification and a learned score-level ensemble, finding that fusing zero-shot DINOv3 and SAM maps *lowered* fine-tuned DINOv2 accuracy by 6.75 pp.
- Transferred the pipeline to articulated animals on AP-10K (807 pairs, 10 species) with multi-layer features, PCA whitening and MNN, reaching 68.1% PCK@0.10 under strict diagonal normalisation.

Subgroup-Disaggregated Faithfulness in Explainable AI *XAI · Fairness · Trustworthy ML*
Code ↗

- Designed a fairness-aware framework for evaluating explanation faithfulness across demographic subgroups, introducing **Worst Subgroup Faithfulness (WSF)**, **Faithfulness Disparity (FD)** and **Explanation Desert Index (EDI)**.
- Evaluated SHAP and LIME on a Random Forest trained on the Adult Income dataset, using Prediction Gap on Important (PGI) and Unimportant (PGU) features disaggregated by sex, race and age.
- Showed that SHAP's stronger global faithfulness (PGI 0.159 vs 0.115 for LIME) masked large subgroup gaps: faithfulness for women was about one-fifth of that for men (0.044 vs 0.208) and negative for under-25s, with disparity above the 0.1 concern threshold by sex and age for both explainers.
- Released a reproducible pipeline (Pandas, scikit-learn, SHAP, LIME) with an interactive Streamlit demo.

Mistake Detection in Procedural Videos *Video Understanding · Temporal Modelling*
Advanced Machine Learning, Politecnico di Torino

- Built step-level mistake detection and task-correctness verification from pre-extracted video features, comparing MLP, LSTM and Transformer temporal models.
- Extended the approach to recipe-level verification from sequences of step embeddings; temporal models outperformed non-temporal baselines.

RESEARCH & PROJECT EXPERIENCE

Industry Challenge Project (8 ECTS) – Agentic Research Assistant Sep 2025 – Feb 2026
Reply × Politecnico di Torino, Turin, Italy **Code** ↗

- Designed and implemented an agentic assistant for question answering over scientific papers, on a brief set by Reply; mentored by one Reply mentor and five Politecnico di Torino mentors.
- Built a coordinator agent for query decomposition and routing, an evidence agent (GROBID citation extraction, arXiv retrieval) and a reasoning agent that cites paragraph- and line-level source locations, plus a feedback loop routing user corrections back to the relevant agent. Stack: TypeScript, LangGraph, OpenAI Agents SDK, Express.

Member, MALTO – Machine Learning Student Team Feb 2026 – Present
Politecnico di Torino, Turin, Italy **Code** ↗

- **3rd of 43** in the MALTO Recruitment Hackathon (Kaggle): fine-tuned Qwen 2.5 with QLoRA on an A100 GPU for multi-class text classification (0.979 validation macro-F1).
- Since October 2026, working with the team on three SemEval-2027 shared tasks.

Multi-Agent Financial Fraud Detection – Reply AI-Agent Hackathon Apr 2026
Team of 3 **Code** ↗

- LangGraph pipeline for business-email-compromise and payment-redirect fraud: a deterministic 15-feature risk scorer feeds parallel transaction and communication agents (GPT-4o-mini, GPT-4o) and a supervisor agent issuing FRAUD / REVIEW / CLEAN verdicts, with thresholds calibrated to maximise F1 under a minimum-recall rule.

Research Assistant Feb 2023 – Jun 2024
Islamic Azad University, Central Tehran Branch, Tehran, Iran

- Personality recognition from text using Transformer-based encoders and hybrid CNN/LSTM architectures; studied contextual language representations and sequential modelling for NLP classification.

Research Intern Jul 2022 – Aug 2022
MECO – MAPNA Electric & Control, Iran: LoRaWAN protocol design and deployment for low-power IoT devices

TECHNICAL SKILLS

Deep learning	PyTorch, Hugging Face Transformers, PEFT (LoRA, QLoRA), TensorFlow/Keras, scikit-learn
Computer vision	YOLOv8-seg, DINOv2/v3, SAM, UNI, H0-mini, Macenko stain normalisation
Analysis & XAI	SHAP, LIME, UMAP, t-SNE, K-Means, HDBSCAN, NumPy, Pandas
LLM & agentic AI	LangGraph, LangChain, OpenAI Agents SDK, RAG, multi-agent systems, GROBID, Langfuse
Programming	Python, TypeScript, JavaScript, Java, SQL, Linux, Git, pytest, Streamlit, Spark, Hadoop

TEACHING

Teaching Assistant, Islamic Azad University, Central Tehran Branch Sep 2022 – Jun 2023
Linear Algebra (Dr Payam Rabanifar, Dr Roohollah Barzamini); Electric Machines I (Dr Payam Rabanifar)

Laboratory Assistant, Islamic Azad University, Central Tehran Branch Feb 2022 – Jul 2022
Telecommunication Circuits Laboratory, under Dr Leila Ghabeli

LANGUAGES & REFERENCES

Persian: native • **English:** C1 (IELTS 7.0, 2024) • **Italian:** A2

References available on request. Open-source code for the projects above: github.com/Hesam-AHT.